

MTH310/MTH520: Graph Theory (GT)

Quiz 3

April 4, 2025

Solve all questions in your answer booklet. **There is no choice this time!**

Question 1.1. [CO3] A k -factor of a graph G is a k -regular spanning subgraph of G and a k -factorization of G (alternatively known as a k -factor decomposition of G) is a set of k -factors $H_1, H_2, \dots, H_\ell$, such that $E(H_i) \cap E(H_j) = \emptyset, \forall 1 \leq i \neq j \leq \ell$ and $\cup_{j=1}^\ell E(H_j) = E(G)$ (and if such a factorization exists for G , G is said to be k -factorable)

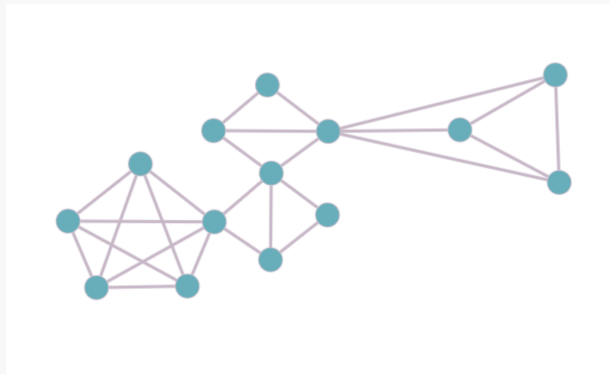
In the context of matching, a 1-factor is almost but not the exact same as a perfect matching M . The corresponding 1-factor of M will be $(V(G), M)$ and a 1-factorization of G will be a partition of its edge set into 1-factors (perfect matchings). Prove the following statements regarding graph 1-factorization:

- a) Prove that if G is k -factorable, G is ℓ -regular where k divides ℓ .
- b) Show that every regular bipartite graph is necessarily 1-factorable.
- c) For which $n \in \mathbb{N}$, is K_n 1-factorable?
- d) Prove that G is 1-factorable implies that G has no cut vertex.

[1+1+1+2 = 5 points]

Question 1.2. [CO2] We can use the blocks of a graph G to develop a technique to find $\tau(G)$ for general connected G :

- a) For any graph G , prove that $\tau(G) = \prod_{B \text{ is a block of } G} \tau(B)$
- b) Compute the number of labelled spanning trees for the following graph:



[3+2 = 5 points]